



University of Asia Pacific

CSE 302: Object Oriented Programming II Lab: Visual and Web Design
Project Documentation

BookLoop Library

Submitted to
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Submitted by

Group Name: Status 200 OK

Group Number:

Name	Registration ID
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Introduction

Project Description:

This project is a digital library management system where users can borrow books by temporarily paying their value, which is refunded upon return. Users can also write reviews after returning books to help others choose better. The admin can manage the system efficiently by adding new books and categories.

Project Motivation:

The motivation is to create a responsible and engaging library system that prevents book loss and promotes reading. By ensuring transparency, accountability, and user interaction, this project modernizes the traditional borrowing process and encourages a community-driven reading culture.

Class Diagram



Book

Represents a book in the library.

- FKs: **added_by** → who added the book; **category** → related book categories.
-

Borrow_record

Tracks each book borrowing activity.

- FKs: **user** → borrower; **book** → borrowed book.
-

Category

Defines book genres or groups.

- FK: **added_by** → who created the category.
-

Discussion

Stores user comments on books.

- FKs: **user** → commenter; **book** → discussed book.
-

Review

Holds user reviews and ratings.

- FKs: **user** → reviewer; **book** → reviewed book; **borrow_record** → related borrow instance.
-

Transaction

Logs user payments and deposits.

- FK: **user** → who made the transaction.
-

User

Extends Django's base user with library info.

- FKs: **user** → base Django user; **borrowed_books** → books currently borrowed.

Project Structure:

1. Books App:

- Purpose:
- Models:
 - Book
 - title
 - description
 - author
 - add_date
 - borrow_price
 - book_image
 - added_by
 - category
 - quantity
 - total_borrowed_time

2. Borrow_records App:

- Purpose:

- Models:

- Borrow_record

- user
 - book
 - borrow_date
 - return_status
 - due_date
 - return_date
 - created_at

3. Categories App:

- Purpose:

- Models:

- category

4. Discussions App:

- Purpose:

- Models:

- Discussion

5. Reviews App:

- Purpose:

- Models:

- Review

6. Transactions App:

- Purpose:
- Models:
 - Transaction

7. Users App:

- Purpose:
- Models:
 - User

Installation and Setup:

- 1. Clone the Repository:**
 - Clone the project repository from GitHub to your local machine.
- 2. Navigate to the Project Directory:**
 - Using your command line interface, navigate to the directory where you cloned the project.
- 3. Create and Activate Virtual Environment (Optional but Recommended):**
 - Create a virtual environment to isolate project dependencies. This step helps to avoid conflicts with other projects.
 - Activate the virtual environment to enable isolation of dependencies.
- 4. Install Dependencies:**
 - Install project dependencies using pip. The dependencies are listed in a **requirements.txt** file in the project.
 - Use the command **pip install -r requirements.txt** to install all dependencies listed in the file.
- 5. Set Up Environment Variables:**
 - Create a **.env** file in the project directory to store environment variables.
 - Define required environment variables such as database connection settings, secret key, etc.
 - Ensure the **.env** file is correctly configured with the necessary variables.
- 6. Run Database Migrations:**
 - Apply database migrations to create the necessary database schema.

	• Use the command python manage.py makemigrations && python manage.py migrate to run migrations and create the database schema.
7. Create Superuser:	• Create a superuser to access the Django admin interface. • Use the command python manage.py createsuperuser and follow the prompts to create a superuser account.
8. Run the Development Server:	• Start the development server to run the project locally. • Use the command python manage.py runserver to start the development server.
9. Access the Application:	• Open a web browser and navigate to the URL provided by the development server (usually <code>http://localhost:8000</code>) to access the locally running application.

Contribution

- github link-----
- screenshot-----

Table 1: Contribution of each member

SL	Full Name	Reg ID	Github username	Number of Commits	No of Lines added	No of Lines deleted
1.	Nafis Fuyad Niloy	22201223	NFNiloy			
2.	Md. Jubaer	23101008	rafnanjubaer			
3.	Md. Osman Goni	23101012	Water-Pot			
4.	Aeysha Tabassum	23101014	Aeysha14			
5.	Arannamoy Mondal	23101024	Arannamoy-Mondal			

Project Summary Table

Components	Required	This project	Summery
Number of App	3	7	Name of the apps: users,books,categories,borrow_records,transactions,discussions,reviews
Number of Class	7	7	Name of the classes: User,Book,Category,Borrow_record,Transaction,Discussion,Review
Number of foreign key	2	11	Foreign keys are used in the following 6 classes and they are - Book,Category,Borrow_record,Transaction,Discussion,Review
Use of Media	2	4	Media is used in the following 4 classes and they are - User,Book,Discussion,Review

CEP Justification Table

Aspect	Attribute	Justification for Django E-commerce Project	CLO	PLO
Knowledge (K)	K2 – Mathematics, numerical analysis, information science	Required for fine-tuning database queries, calculating book borrow prices, and handling user transactions accurately.	CLO1	PLO1
	K3 – Engineering fundamentals	Understanding web architecture, database normalization, and data relationships between users, books, and transactions.	CLO1	PLO1
	K4 – Specialist knowledge	Application of Django ORM, secure SSL payment gateway integration, and relational database design for book borrowing system.	CLO1, CLO3	PLO1, PLO5
	K6 – Knowledge of practice	Deployment on cloud (e.g., Firebase/Netlify), managing APIs, and maintaining system updates through CI/CD practices.	CLO2, CLO3	PLO3, PLO5
	K7 – Role of engineering in society	Promotes digital literacy, fair access to educational resources, and ensures ethical data handling for users.	CLO4	PLO6
Problem (P)	P1 (mandatory): Requires in-depth engineering knowledge	Involves integration of user authentication, transactions, category mapping, and database relations using multiple knowledge areas (K2–K7).	CLO1	PLO1
	P2: Conflicting requirements	Trade-offs between fast data access and secure payment/storage; optimizing book search speed vs. server cost.	CLO2	PLO2

	P3: No obvious solution	Designing dynamic borrow/return flow, handling overdue penalties, and preventing double borrowing requires creative logic.	CLO2, CLO7	PLO2, PLO3, PLO10
	P6: Multiple stakeholders	Students, librarians, administrators, and payment gateway providers interact through different modules.	CLO4, CLO6	PLO6, PLO9
	P7: Interdependent parts	Book, Borrow_record, Review, Discussion, and Transaction systems are tightly linked and depend on one another.	CLO2	PLO3
Activities (A)	A1: Diverse resources	Requires cloud hosting, database servers, Django framework, SSL payment API, and admin/user collaboration.	CLO2, CLO6	PLO3, PLO9
	A2: Conflicting issues	Balancing data security, payment accuracy, and quick response for borrowing and returning operations.	CLO2, CLO4	PLO2, PLO6
	A3: Creative use of knowledge	Applying analytics to track borrowing trends, predict popular books, and enhance user experience with recommendations.	CLO2, CLO7	PLO3, PLO10

Screenshot

BookLoop

Browse Books

Login

Create your Account

USERNAME

FIRST NAME

EMAIL ADDRESS

CONTACT NO

GENDER

PASSWORD

PASSWORD CONFIRMATION

October 2025

S M T W T F S

28 29 30 1 2 3 4

5 6 7 8 9 10 11

12 13 14 15 16 17 18

19 20 21 22 23 24 25

26 27 28 29 30 31 1

2 3 4 5 6 7 8

Clear Today

mm/dd/yyyy

Sign Up

Sign Up

BookLoop

Browse Books

Login

Log in to your account

Or sign up for a new account

Username

user1

Password

Log In

Copyright © 2025 - All right reserved by Status 200 Ok

Login

BookLoop

Browse Books

Borrow History

My Review

Profile

Deposit

Transaction History

1000.00 BDT

Logout

Please deposit first.

Amount

5000

Deposit

BookLoop

Browse Books

Borrow History

My Review

Profile

Deposit

Transaction History

6000.00 BDT

Logout

OREILLY

Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow

Concepts, Tools, and Techniques to Build Intelligent Systems

powered by

Aurélien Géron

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems

Author: Aurélien Géron

Handpicked books for you.

Total Borrowed Time: 1

View Details

Welcome to BookLoop

Discover, borrow, and enjoy thousands of books

Filter & Search Options

DEMO

Support

FAQ

Offers

Login

CARDS

MOBILE BANKING

NET BANKING

VISA

Other Cards

Enter Card Number

First digit is 37 or 4 or 5 and rest digits are 1

MM/YY

CVC/CVV

Card Holder Name

Save card & remember me

By checking this box you agree to the [Terms of Service](#)

PAY 5,000 BDT

OTP Page

Do not press browser back or forward button while you are in payment page

Payment Summary

Please review the following detail for this transaction:

Amount: 5000.00

Invoice number: 2510102015050dakBGYDzPZ8Md0

Description: Products

Enter Card Information

OTP:

Success

Success with risk

Your entered card information could not be corrupted or become known to the third party, as all transmitted data is encrypted by the SSL protocol.

Note

1. For VISA and MC, look at the back side of your Card to find 3-digit CVV2/ CVC2.

For AMEX, look at the upper right corner of the front side of your Card to find 4-digit CSC.

2. The cardholder's name should be entered just as it's written on the card.

Verified by VISA

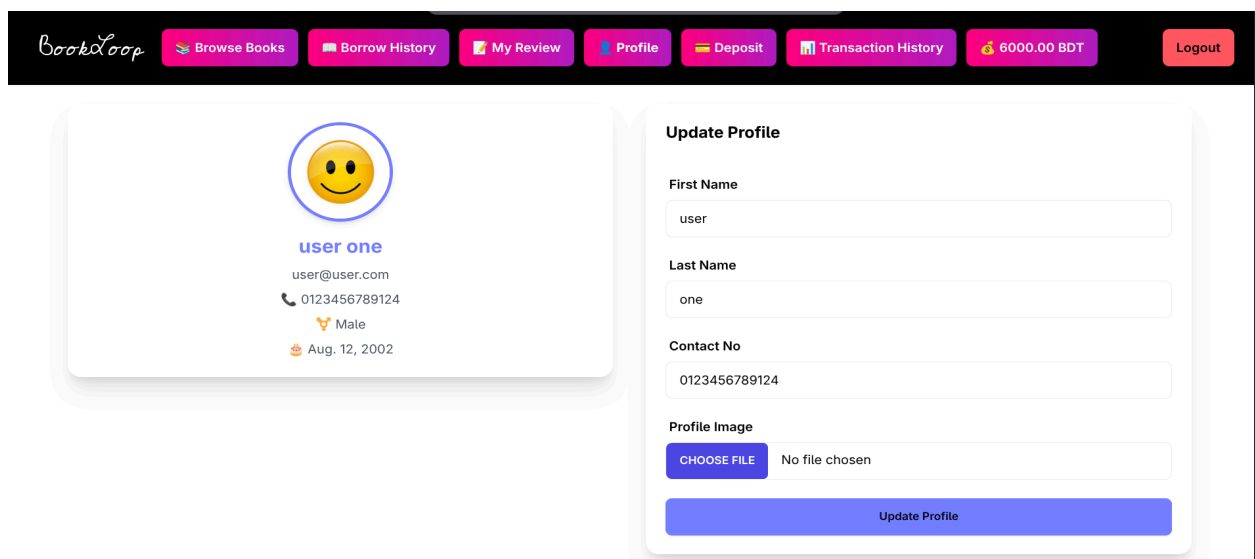
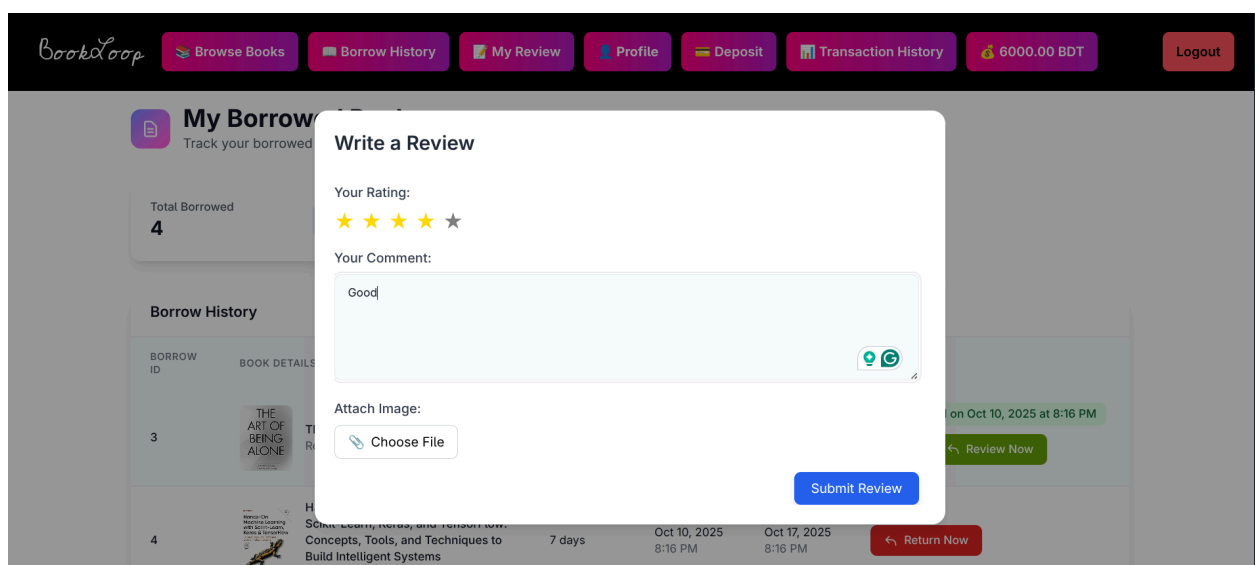
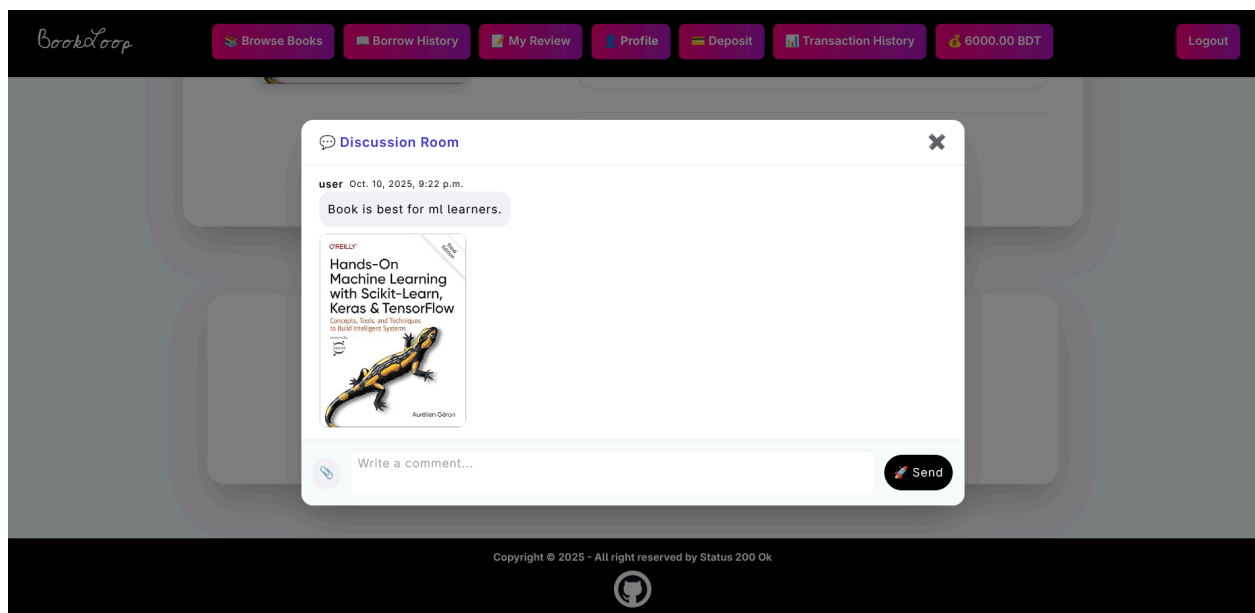
AMERICAN EXPRESS SafeKey

MasterCard SecureCode

SSLCOMMERZ TESTBOX GATEWAY (NO CARD INFORMATION WILL BE SAVED AND DUMMY CARD WILL BE DISPLAYED IN RESPONSE)

TRX ID	TYPE	AMOUNT	STATUS	TIME	REFERENCE
#8	Credit	\$5000.0	Success	Oct. 10, 2025, 8:15 p.m.	TXN27
#7	Credit	\$620.0	Success	Oct. 10, 2025, 5:42 p.m.	TXN26
#6	Debit	\$120.0	Success	Oct. 10, 2025, 4:54 p.m.	TXN2BB5
#5	Credit	\$120.0	Success	Oct. 10, 2025, 4:47 p.m.	TXN2RB4

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- Attach screenshot of key features along with short explanation.

Attach screenshot of key features along with short explanation.